

INTELLIGENT HEALTHCARE MANAGEMENT: AN INTEGRATED SYSTEM COMBINING AI-POWERED APPOINTMENT SCHEDULING AND MEDICINE MANAGEMENT

A PROJECT REPORT

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BONAFIDE CERTIFICATE

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ABSTRACT

Managing health after a consultation is a fragmented task. Patients book appointments in one place, carry a paper or photographed prescription, enter medicines manually into another application, and track reminders, adherence and refills separately. This project presents PillSync, an intelligent healthcare management system that joins these activities in a single platform. It is built with a React frontend, a FastAPI backend, REST APIs and a PostgreSQL database.

The system lets a patient schedule an appointment and, after the consultation, upload a prescription image. Optical Character Recognition extracts the medicine name, strength, dosage and frequency and converts them into structured medicine records that the patient can review and save. These records drive a daily medication schedule with reminders, and each dose can be recorded as taken, missed or snoozed.

The recorded activity is converted into adherence analytics, while drug-interaction detection warns about potentially conflicting medicines and inventory monitoring supports timely refills. Result screens of every module are presented and discussed. By linking appointment scheduling, prescription digitization and continuous medication support, the system reduces manual effort and offers a more organised, patient-centred healthcare experience.

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LIST OF ABBREVIATIONS

ABBREVIATION	EXPANSION
AI	Artificial Intelligence
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
DB	Database
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
OCR	Optical Character Recognition
REST	Representational State Transfer
SQL	Structured Query Language
UI	User Interface
UX	User Experience

CHAPTER 1

INTRODUCTION

1.1 GENERAL

The process of receiving treatment isn't finished when a patient steps out of the doctor's office. Most of the hard work begins later; the patient has to understand the prescription, buy medicines, take the medicines regularly at the right hour, and ensure that the treatment process is not disturbed. In practice, all these processes are done using separate means. Booking the doctor's appointment through a specific application or a website is one way, while carrying a paper or the picture of the prescription is a separate process. Entering the prescribed medications in the reminder app on paper is something completely different, and tracking refill dates is also a separate process.

The current project titled **Intelligent Healthcare Management** unites all these functions on one web platform. The PillSync application allows the patient to book an appointment, send a picture of the prescription to the application, and automatically receive medicine specifications using OCR technology. The received data is converted into a medicine record that performs the function of an automatic daily reminder, schedule, adherence report, and drug interaction warning.

The fundamental concept is continuity. Instead of considering appointment scheduling, prescription management, and follow-up as three independent processes, the system considers them as subsequent phases of a patient journey. The information gained in one phase is put to use in the next. Thus, a patient inputs less data, minimizes errors, and has a single dashboard to monitor the progress of treatment.

From a technological perspective, the platform uses a conventional three-tier architecture with React as the front-end component, FastAPI as the back-end one, and PostgreSQL database integrated via REST APIs. Intelligent technologies are utilized to avoid unnecessary manual input: in making appointment allocations, interpreting prescription images, finding unsafe medicine combinations, and predicting the need for refills.

1.2 PROBLEM STATEMENT

- The issue being addressed in this framework is fragmentation. At each level of outpatient care, different instruments are typically employed and there aren't any links between them. The problems observed can be arranged as follows.

- **Appointments.** Patients have difficulty locating appropriate services and free reception hours, tracking their visits – upcoming and cancelled. Scheduling systems rarely have information on patients' whereabouts after the consultations.
- **Prescriptions.** Prescriptions are often either handwritten or photographed. The medicine name, dose, mode of intake, frequency and instructions have to be read and recorded, which is a slow procedure with high risk of making mistakes.
- **Medication management.** If a patient is taking several medications, he/she needs to memorize different doses and schedules while being aware of any shortages.
- **Adherence.** Doses are forgotten, delayed or missed, and patients seldom have a clear understanding of how compliant they have really been [1], [2].
- **Safety and supply.** Combinations of medicines may interact, and stock levels are rarely monitored until a medicine has already finished.

A single platform that joins these stages, reduces manual entry and keeps the patient informed after the consultation is therefore needed.

1.3 OBJECTIVES

The main intention is to set up an integrated system that would provide an easy way to arrange appointments and keep records of prescriptions and any ongoing treatment plan. The goals are specified as follows:

- Enabling patients to book an appointment conveniently by having the options of selecting the date, time, type of consultation and reason for the visit while keeping the central database of appointments.
- Using OCR technology to transform paper prescriptions into the electronic format, which would extract the medicine name and strength, dosage, frequency and instructions from an image of a prescription.
- Transforming the text obtained from the OCR into the electronic format of medicine records for patients to view, change and store them.
- Giving the patients opportunities to enter their records of prescribed medicine to check their amount or dosage or frequency.
- Making an everyday schedule of taking medicine and creating reminders for every taken dose.
- Translating dose information into the data about adherence to treatment.
- To alert patients about certain combinations of drugs that may not be safe.

- To keep track of available medicine stock level and help with the refill process whenever necessary.
- To decrease physical work in healthcare delivery and to provide the patient with individual healthcare process.

1.4 EXISTING SYSTEM

Current solutions are indeed helpful, but their scope is rather limited. The online appointment systems only provide availability and confirmation options that leave them out of the process after the appointment is booked. As for prescriptions, they are still mainly handled manually, so it is usually up to the patient or their family to interpret either a photographed or handwritten prescription and rewrite it. Generic reminder apps are capable of notifying the user about the appointment at the designated time, but they also rely on the fact that the user has entered all data correctly initially.

There are stand-alone tools for interaction checking, adherence monitoring, and stock control, but each of them is operating in its own vacuum. This means that the patient who needs a full package of services has to download multiple apps, enter the same data of medications over and over again, and reconcile the results in his/her mind.

1.5 PROPOSED SYSTEM

The proposed approach is a single web application where all modules share the same data. A patient can book an appointment. After finishing the appointment, a patient can take a picture of the prescription and upload it; the Optical Character Recognition module will read it and create a table of medicines, their strength, dosage, and schedule. When a patient confirms the table, all data get saved in medicine module and become the basis for future actions.

The medicine list builds a timeline of doses and sends reminders. A patient can mark each dose as taken, missed, or snoozed; all of those will appear in the adherence report. In addition, the system checks the medicine list for interactions and compares the quantities of medications available with the intake.

The entire system is constructed around FastAPI and PostgreSQL, which means all information produced in one module immediately becomes available to others. Therefore, the major advantage of this solution is not any special feature but rather establishing the workflow from the appointment to refill of medicines.

CHAPTER 2

LITERATURE SURVEY

2.1 LITERATURE REVIEW

Each of the areas dealt with in this project has been examined individually. The current chapter addresses these areas one by one and then describes the gap that the present study tries to fill.

2.1.1 Online Appointment Scheduling

Appointment systems found on the internet relieve the pressure from making phone calls and minimize the struggle to sit and wait. Most of the work in this field is related to the availability of slots, canceling appointments and handling work of the providers. Most of the work ends with confirmation of the booking but information does not ordinarily flow to the medication part of treatment after the consultation.

2.1.2 AI-Assisted Appointment Allocation

Recently introduced systems utilize data-driven methods for appointment allocation according to availability of providers, estimated length of consultation and patient details to ensure even distribution of slots. This project appointment module works in a similar way as it utilizes available slots along with client and provider details for booking.

2.1.3 Prescription Digitization and OCR

The process of optical character recognition (OCR) allows copied images of characters to be converted into characters that the machine recognizes due to open-source engines like Tesseract [3]. Printed characters fare well in recognition, but prescriptions are harder to decipher due to uneven photographs, stamps and abbreviations. Research on digitization of prescriptions thus combines pre-processing of the image, extraction of the text as well as a learned or a rule-based process to tell apart the segments of text that refer to names of medicines, their strength and frequency. The same sequence of pre-processing, recognition and field identification is utilized in this project.

2.1.4 Medication Management and Reminder Applications

With mobile and web reminder applications, patients are able to adhere to a medication regimen. However, the effectiveness of these applications is determined by the quality of the data entered, and most of these applications require the patient to enter their medication, dose and time manually.

Moreover, these applications are typically not related to the prescription that originally prescribed the medication.

2.1.5 Medication Adherence Monitoring

Noncompliance is an established issue in maintenance therapy and trade organizations have demonstrated that it is common for patients to take less medication than what has been prescribed [1], [2]. Research differentiates between whether a patient starts a complicated therapy, whether there is consistency in taking the prescribed treatment, and how long an individual continues the therapy [4]. Keeping track of how many doses have been taken, missed, or deferred during the study provides all the information necessary to present such consistency in a statistical way and in charts.

2.1.6 Drug-Interaction Detection

Taking certain medications at the same time may be ineffective or may worsen the likelihood of experiencing adverse side effects. Interaction checkers take the patient's medication list and compare it to a database of known interactions and give warnings. In a healthcare management solution, this check is particularly valuable when it occurs automatically using the patient's own medications.

2.1.7 Refill and Inventory Management

Not having enough medicines is one of the most common reasons why treatments are interrupted. Inventory tracking and refill prediction determine when the stock will run out based on the current stock and dosing frequency and notify the patient about such situation in advance. Only a few consumer tools combine this feature with prescription information and reminders.

2.2 COMPARISON AND DISCUSSION

Table 2.1 summarises how the main categories of existing tools compare with the proposed platform.

Table 2.1 Comparison of Related Approaches

Approach	Main Focus	Strength	Limitation
Online appointment portals	Slot booking and confirmation	Convenient scheduling, central records	No link to prescriptions or medicines
Prescription OCR tools	Text extraction from images	Reduces typing effort	Output is not connected to schedules or reminders
Reminder applications	Dose alarms	Simple and widely used	Manual data entry; limited analytics
Interaction checkers	Medicine safety	Warn about conflicts	Stand-alone lookup, separate data

Approach	Main Focus	Strength	Limitation
Refill / inventory trackers	Stock monitoring	Help avoid shortages	Not tied to adherence or prescriptions
Proposed PillSync platform	End-to-end patient workflow	All of the above on one shared data store	Accuracy depends on prescription image quality

According to the comparison, every individual capability has been implemented somewhere before; what is not implemented is the connection between capabilities. A patient who utilizes OCR is still required to transfer the result to another application, and a patient whose adherence diagrams have already been created has to enter the doses manually. Filling in these missing links is the aim of this research.

2.3 CONCLUSION

Research proves that scheduling, prescription OCR, adherence monitoring, interaction checking and inventory management have been advancing progressively. Therefore, the review calls for a system that integrates all these systems into one patient-centric workflow, which is further discussed in this section.

CHAPTER 3

SYSTEM DESIGN

This chapter explains how PillSync has been designed before any of its modules are described in detail. It covers the architecture, the requirements the system must satisfy, the use cases, the data model, the flow of data and the overall workflow that links the modules. The design decisions follow one principle: information entered or produced once should be reused by every later stage of the patient journey.

3.1 SYSTEM ARCHITECTURE

PillSync follows a frontend–backend–database architecture. The patient interacts only with the React frontend, which presents the dashboard and forms. Every action is sent to the FastAPI backend as a REST request, and the backend applies the application logic and reads or writes data in the PostgreSQL database [5]-[8]. Figure 3.1 shows the overall structure.

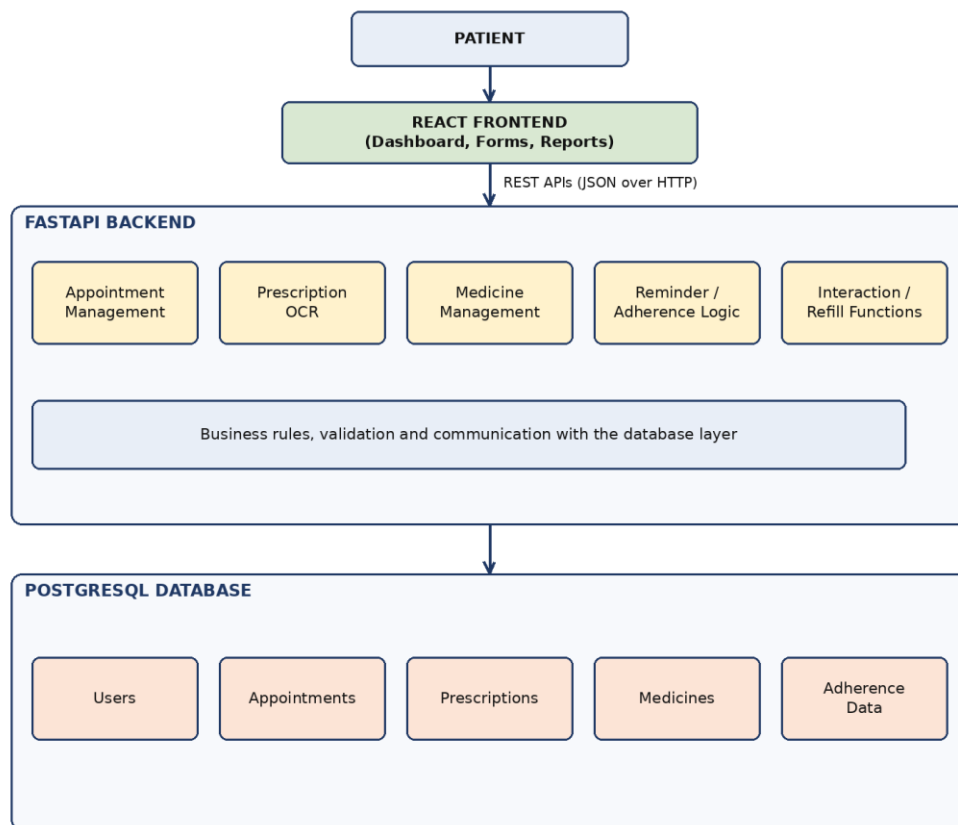


Figure 3.1 Overall System Architecture

The backend is divided into five logical parts: appointment management, prescription OCR, medicine management, reminder and adherence logic, and interaction and refill functions. Because all of them share one database, the output of one module can be consumed by another without any manual transfer. For example, a prescription processed by the OCR part appears as

medicine records in the medicine part, and from there in the reminder, interaction and inventory parts.

3.1.1 Presentation Layer

The presentation layer is the React app that is running inside the patient's browser. This layer is in charge of everything the patient sees, including the side menu, the appointment booking window, prescription table, schedule, compliance diagram, and inventory cards. The presentation layer has no business rules of its own and only collects information and delivers it to the back end and displays the back end results. It makes it easier to change the look of the app, as changing the presentation layer does not affect the layers that incorporate the business logic.

3.1.2 Application Layer

The application layer is the FastAPI backend. It receives each request, checks that the data is valid, applies the rules of the relevant module and then talks to the database. Every module is kept separate inside this layer, so the OCR code does not need to know how the adherence report is calculated and the inventory code does not need to know how slots are allocated. When one module needs data from another, it asks for it through the shared database rather than calling the other module directly.

3.1.3 Data Layer

PostgreSQL relational database serves as data layer for this project. It contains information about users, appointments, prescriptions, medicines and adherence, and some data regarding the relationships between them. The data is kept in the relational store because the data is inherently interconnected: each patient is assigned multiple appointments, each prescription consists of several medicines and each medicine has many doses.

3.1.4 Communication Between Layers

The frontend and backend exchange data as JSON over HTTP using REST principles [5]. Each module exposes a small set of operations, such as creating an appointment or recording a dose, and the frontend calls these operations when the patient presses a button. Because the interface between the layers is a plain REST contract, either side can be improved independently as long as the contract is respected.

3.2 DESIGN CONSIDERATIONS

Healthcare software has to be dependable in ways that a casual application does not. The following goals guided the design of PillSync. Table 3.1 lists them together with the way the design addresses each one.

Table 3.1 Design Goals and Design Response

Design goal	Meaning for PillSync	Design response
Continuity	Data created in one stage should be reusable in the next	One shared relational database for all modules
Accuracy	Medicine details must be correct before they drive reminders	OCR output is shown in an editable table and saved only after confirmation
Safety	Unsafe combinations of medicines should not go unnoticed	Interaction analysis runs on the stored medicine list and displays warnings
Usability	Patients, including elderly users, must find the system easy	Simple navigation, large action buttons, one-click dose actions
Modularity	Features should be changeable without breaking others	Separate backend modules behind a common REST interface
Extensibility	New features such as notifications should be easy to add	Layered architecture with clear module boundaries
Reliability	Records must not be lost or become inconsistent	Relational integrity through primary and foreign keys

3.3 SYSTEM REQUIREMENTS

3.3.1 Functional Requirements

The system has things it needs to accomplish. They were derived from the objectives in Chapter 1 and are summarised in Table 3.2.

Table 3.2 Functional Requirements

ID	Requirement	Module
FR1	The system shall allow a patient to view available slots and book an appointment	Appointment
FR2	The system shall store appointments with a status of scheduled, completed or cancelled	Appointment
FR3	The system shall accept a prescription image uploaded by the patient	Prescription OCR
FR4	The system shall extract medicine name, strength, dosage and frequency from the image	Prescription OCR
FR5	The system shall display the extracted data in an editable form before saving	Prescription OCR
FR6	The system shall allow medicine records to be added, viewed, modified and removed	Medicine
FR7	The system shall generate a daily schedule from the stored medicines	Scheduling
FR8	The system shall let the patient mark each dose as taken, missed or snoozed	Reminders
FR9	The system shall compute adherence statistics from the recorded dose actions	Adherence

ID	Requirement	Module
FR10	The system shall analyse the medicine list and warn about potential interactions	Interaction
FR11	The system shall display current stock and flag low or exhausted medicines	Inventory
FR12	The system shall allow a refill to be recorded for a medicine	Inventory

3.3.2 Non-Functional Requirements

Table 3.3 Non-Functional Requirements

Attribute	Requirement
Usability	The interface should be readable and operable with a few clicks for each task
Performance	Common screens should load and respond without noticeable delay on a normal network
Reliability	Saved records should remain consistent after restarts and repeated use
Maintainability	Modules should be separable so that one can be modified without rewriting the rest
Portability	The application should run in any modern web browser on common operating systems
Security	Patient data should be accessible only to the patient who owns it
Scalability	The relational design should allow more patients and records to be added over time

3.3.3 Software Requirements

Table 3.4 Software Requirements

Component	Technology / Requirement
Frontend framework	React
Backend framework	FastAPI (Python)
Database	PostgreSQL
Client-server communication	REST APIs exchanging JSON
Prescription processing	OCR component for text extraction from images
Client software	Any modern web browser
Operating system	Windows, Linux or macOS

3.3.4 Hardware Requirements

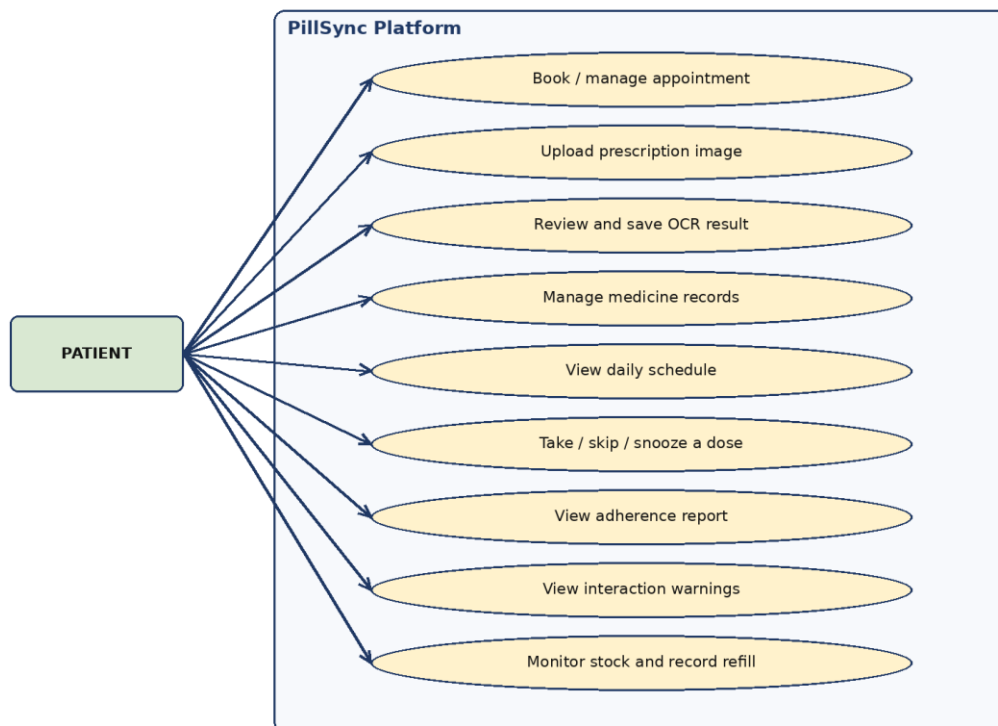
Using this application, one does not need to worry about their computer's specifications much. The setup below is an example of what one needs for using and showing the application.

Table 3.5 Hardware Requirements

Item	Suggested Configuration
Processor	Processor must be dual-core or better Processor 2 GHz or more.
Memory	Memory should be 8 GB RAM, Memory 4 GB minimum.
Hard Drive	Hard Drive should have least 10 GB free disk space, Hard Drive.
Network	Internet or any local network connection
Camera / scanner	For taking prescription images (not obligatory since it could not be uploaded)

3.4 USE CASE VIEW

The primary actor of PillSync is the patient. Figure 3.2 shows the use cases that the patient can perform, and Table 3.6 describes them briefly. Administrators and providers are mentioned in the project description as users who can monitor appointment records; their screens are outside the scope of the implemented results presented in this report.

**Figure 3.2 Use Case Diagram of the Patient****Table 3.6 Actors and Use Cases**

Actor	Use case	Outcome
Patient	Book / manage appointment	Appointment stored with date, slot, type and reason

Actor	Use case	Outcome
Patient	Upload prescription image	Image passed to the OCR module
Patient	Review and save OCR result	Structured medicine records created
Patient	Manage medicine records	Records added, viewed, modified or removed
Patient	View daily schedule	Timeline of today's doses displayed
Patient	Take, skip or snooze a dose	Dose status stored as adherence data
Patient	View adherence report	Statistics and compliance chart displayed
Patient	View interaction warnings	Potential conflicts shown in the safety panel
Patient	Monitor stock and record refill	Quantity updated and low stock highlighted
Provider / administrator	Monitor appointment records	Scheduled, completed and cancelled visits visible

3.5 DATA AND PERSISTENCE DESIGN

PostgreSQL is the relational database. It stores the information that connects one stage of the workflow to the next: users, appointments, prescriptions, medicines and adherence data. Keeping them in one relational store allows a record created early in the workflow, such as a prescription, to be linked to everything derived from it later.

Table 3.7 Core Persistence Entities

Entity	Information Stored	Used By
Users	Patient account and profile details	All modules
Appointments	Date, time slot, appointment type, reason, symptoms and status (scheduled, completed, cancelled)	Appointment management
Prescriptions	Uploaded prescription and the information extracted from it	Prescription OCR
Medicines	Name, quantity, dosage, frequency, instructions, scheduled time	Medicine management, reminders, interaction and refill functions
Adherence data	Outcome of each scheduled dose: taken, missed or snoozed	Adherence analytics

3.5.1 Entity-Relationship Model

Figure 3.3 presents the entity-relationship model. A user owns many appointments and many prescriptions. A prescription may be linked to the appointment during which it was issued, and it contains many medicine records. Each medicine accumulates adherence records over time as doses are taken, missed or snoozed. The attribute lists in the figure are representative of the information described in the project material and are expanded in the tables that follow.

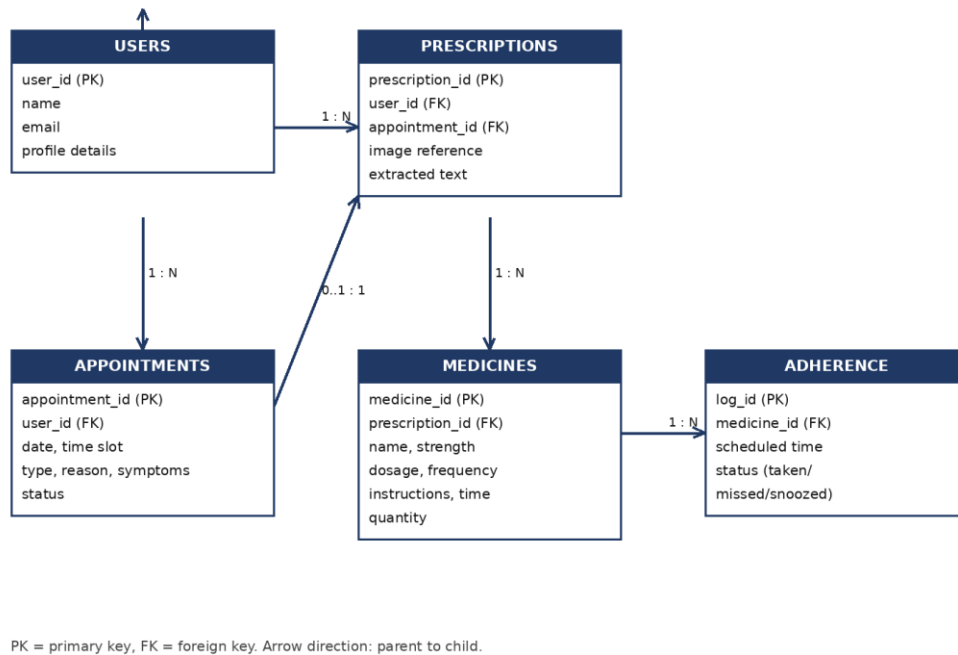


Figure 3.3 Entity-Relationship Diagram

3.5.2 Description of Entities

The following tables describe the attributes of each entity at a logical level. Names of columns are descriptive rather than a transcript of the physical schema.

Table 3.8 Users Entity

Attribute	Description
user_id	Unique identifier of the patient (primary key)
name	Name of the patient
email / login details	Credentials used to access the platform
profile details	Information shown in the profile summary, such as emergency contact

Table 3.9 Appointments Entity

Attribute	Description
appointment_id	identifier for the appointment.
User_id	This refers to the patient who made the booking. It is a key that links, to the user table.
date and time slot	Selected consultation date and the chosen available slot
appointment type	Mode of consultation chosen in the booking dialog
reason / symptoms	Reason for consultation and optional symptoms or questions
status	Scheduled, completed or cancelled

Table 3.10 Prescriptions Entity

Attribute	Description
prescription_id	is the identifier of the prescription
user_id	Owner of the prescription (foreign key)
appointment_id	Related appointment, where applicable (foreign key)
image reference	Reference to the uploaded prescription image
extracted text	Text returned by the OCR step

Table 3.11 Medicines Entity

Attribute	Description
medicine_id	Unique identifier of the medicine record (primary key)
prescription_id	Prescription from which the record was created (foreign key)
name and strength	Medicine name, generic name and strength
dosage and frequency	Amount per dose and how often it is taken
instructions	Additional directions such as timing relative to food
scheduled time	Time(s) of day at which doses are due
quantity	Current stock available with the patient

Table 3.12 Adherence Entity

Attribute	Description
log_id	Unique identifier of the dose record (primary key)
medicine_id	Medicine to which the dose belongs (foreign key)
scheduled time	Time at which the dose was due
status	Taken, missed or snoozed
recorded at	Moment at which the patient responded

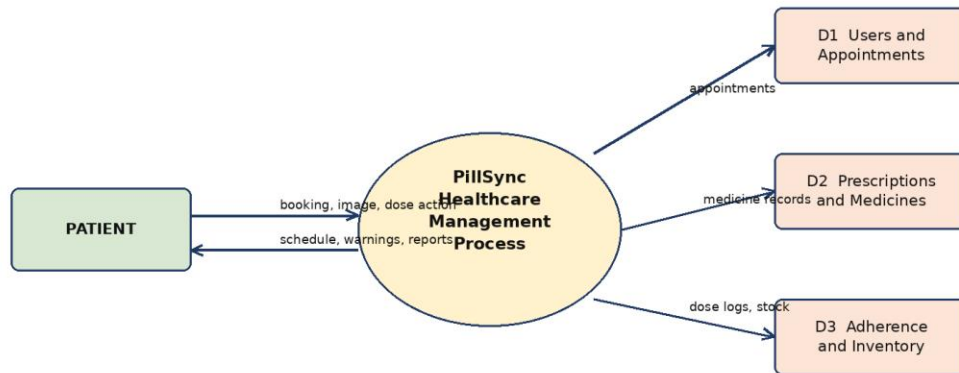
3.5.3 Data Integrity

Relationships between the entities are enforced through primary and foreign keys, which prevents orphan records such as a dose log without a medicine. Because the schedule, adherence report and inventory view are all computed from the same medicine and adherence tables, they cannot disagree with one another. This is one of the practical advantages of a single relational store compared with several independent applications [9].

3.6 DATA FLOW DESIGN

Figure 3.4 illustrates the top-level data flow. The patient gives appointments, prescription photos and drug actions, and gets timetable, alerts and outputs in return. Within the system a

lone central process is exchanging data with three repositories: a user and appointment one, a prescription and medicine one, as well as a compliance and stock repository. The internal modules are hidden in this figure so that one can easily see the data flow on the general level.



Level 0 data flow: one central process exchanges data with the patient and three data stores.

Figure 3.4 Level 0 Data Flow Diagram

3.7 SYSTEM WORKFLOW

The complete patient journey is shown in Figure 3.5. It is divided below into three phases for clarity.

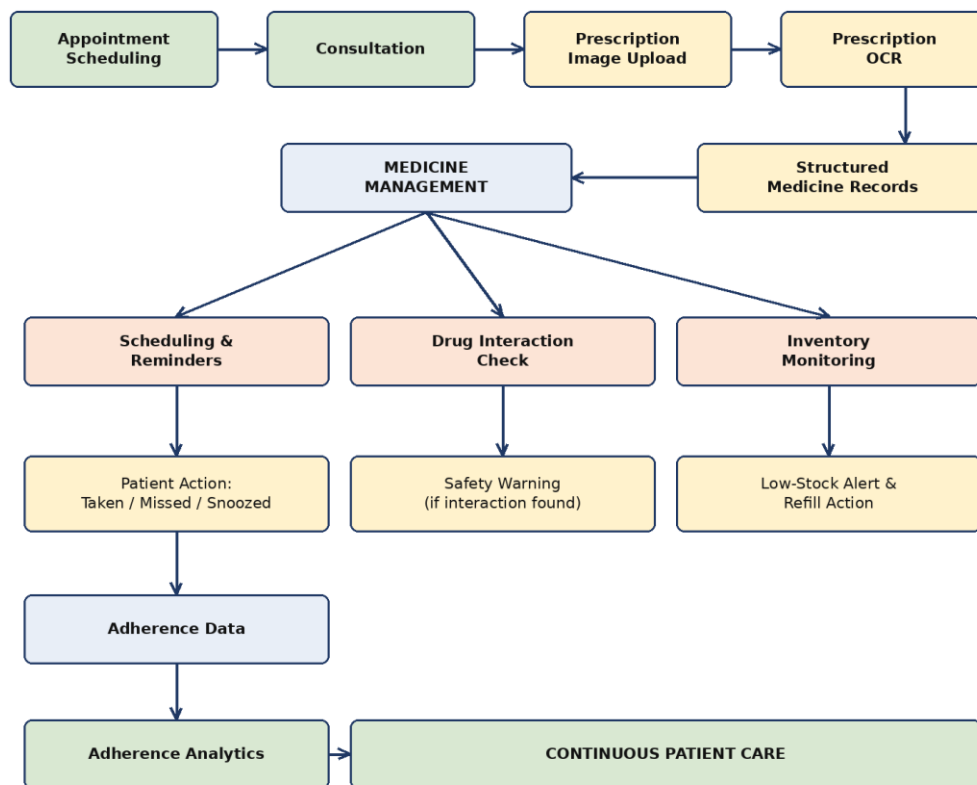


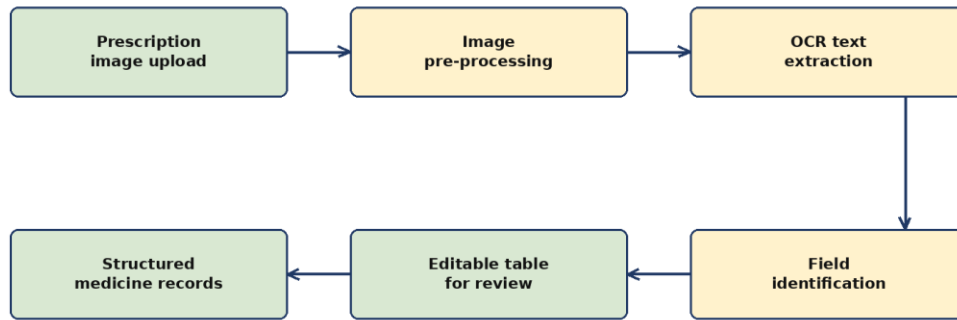
Figure 3.5 End-to-End Healthcare Management Workflow

3.7.1 Appointment Booking and Consultation

The patient browses the available services, selects a provider, views the free slots and submits an appointment request. The appointment is stored with its status and can later be viewed, completed or cancelled. After the consultation the patient receives a prescription.

3.7.2 Prescription Digitization

The patient uploads the prescription image. It is pre-processed and passed through OCR, after which the system identifies medicine names, strengths, dosage and frequency and arranges them in an editable table. The patient verifies the table and saves it, which creates structured medicine records. The stages of this pipeline are shown in Figure 3.6.



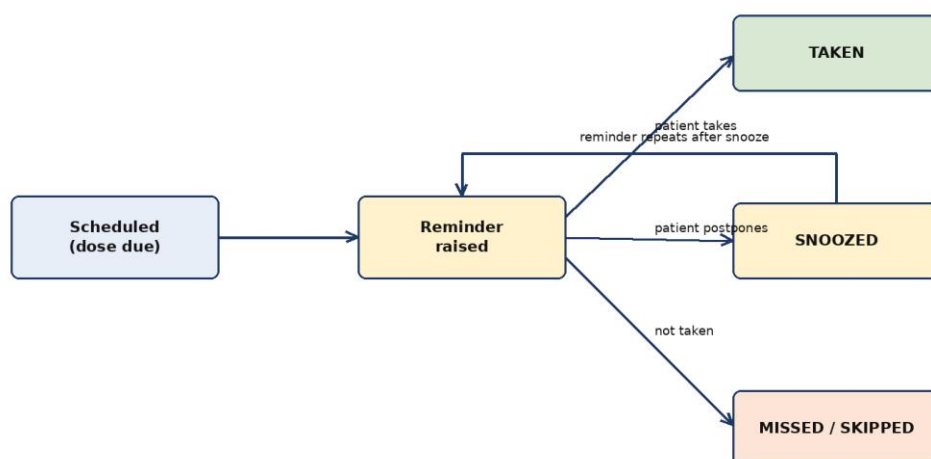
Pre-processing, OCR and field identification are automatic; saving is always confirmed by the patient.

Figure 3.6 Prescription OCR Processing Pipeline

3.7.3 Continuous Medication Support

Saved medicines generate the daily schedule and reminders. Each response from the patient is recorded and accumulates into adherence analytics. Simultaneously the interaction check inspects the combination of medicines and the inventory module compares remaining quantities with requirements, leading to warnings or refill actions when necessary.

Every scheduled dose passes through a small set of states, shown in Figure 3.7. A dose starts as scheduled, a reminder is raised when it becomes due, and the patient then takes it, postpones it or does not take it. A snoozed dose returns to the reminder state after the snooze interval, so it is eventually resolved as taken or missed. Only these resolved outcomes contribute to the adherence figures.



All three outcomes are stored as adherence data and used by the analytics module.

Figure 3.7 State Diagram of a Scheduled Dose

Two further checks run in the background of the medicine list. The first compares the medicines with one another for potential interactions; the second compares the stock with the number of doses still to be taken. Figure 3.8 shows the decision flow of both checks.

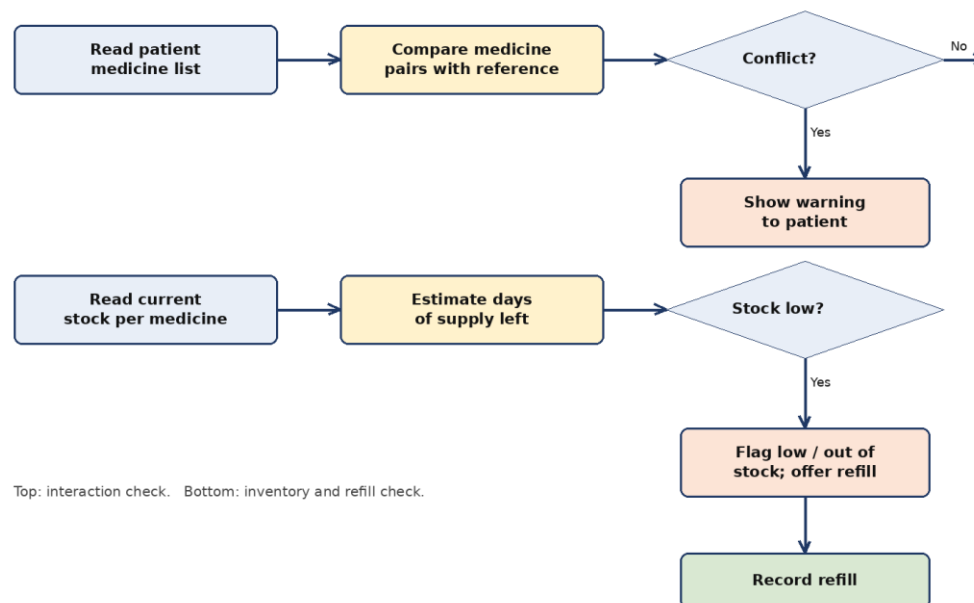


Figure 3.8 Interaction Check and Refill Check Flow

3.8 FUNCTIONAL DESIGN OF BACKEND MODULES

The backend exposes the operations summarised in Table 3.13 to the frontend through REST APIs. The table describes the operations at a functional level.

Table 3.13 Major Functional Operations by Module

Module	Operations
Appointment management	View available slots; create an appointment; list appointments; change status
Prescription OCR	Upload prescription image; extract text; return structured medicine fields; save confirmed records
Medicine management	Add, view, update and delete medicine records
Reminders	Build the daily schedule; raise reminders at scheduled times
Adherence	Record taken / missed / snoozed; compute adherence statistics and breakdown
Interaction and refill	Analyse the medicine list for conflicts; report stock levels; record a refill

3.8.1 REST Resource Organisation

The REST interface groups operations by resource. Table 3.14 shows an illustrative organisation of these resource groups. It describes how the operations are naturally divided and is not a listing of the exact endpoint names used in the source code.

Table 3.14 Illustrative REST Resource Groups

Resource group	Typical operations	Typical caller
Appointments	List slots, create, list, update status	Booking dialog, appointment list
Prescriptions	Upload image, return recognised fields, confirm and save	Prescription recognition screen
Medicines	Create, read, update, delete	Medicines and schedule screen
Schedule and doses	Today's timeline, record dose action	Dashboard timeline
Reports	Adherence summary and breakdown	Reports and analytics screen
Inventory	Stock list, filter, record refill	Inventory and refills screen

3.9 USER INTERFACE DESIGN

The interface follows a dashboard layout with a persistent navigation bar on the left and a content area on the right. The navigation bar lists the main sections of the platform: dashboard, medicines and schedule, appointments, reports, prescriptions, emergency card, notifications and profile. A dark colour scheme with high-contrast accent colours is used so that status information such as warnings, stock levels and adherence figures stands out clearly. Table 3.15 lists the principles that were followed [10].

Table 3.15 User Interface Design Principles

Principle	How it appears in PillSync
Consistency	Same navigation, card style and button style on every screen
Visibility of status	Coloured badges and progress bars show stock and adherence at a glance
Minimal effort	Dose actions are single buttons placed next to each scheduled dose
Error prevention	OCR results are editable and are saved only after confirmation
Recognition over recall	Medicine names, times and instructions are shown together on the timeline

3.10 SECURITY AND PRIVACY

Health Information as sensitive information should take into consideration privacy measures even at the third level. The measures depicted in Table 3.16 pertain to the precautions and considerations that must be implemented during the design phase. They show exactly what the live version of the platform should deliver and how the architecture ensures that everything is fulfilled.

Table 3.16 Security and Privacy

Concern	Consideration	Architectural support
Access control	Each patient should see only their own records	User identifier stored on every record
Data in transit	Communication between browser and server should be encrypted	REST over HTTPS in deployment
Data at rest	Stored data should be protected on the server	Centralised PostgreSQL store with controlled access
Input validation	Uploaded files and form fields must be checked	Validation in the FastAPI layer
Medical responsibility	Warnings are advisory and do not replace a clinician	Warnings displayed to the patient, never applied automatically

3.11 CHAPTER SUMMARY

The system is organised in three layers, uses one relational database as the common memory of all modules and follows a workflow that runs from appointment to refill. The next chapter describes each module in detail, together with the logic used inside it.

CHAPTER 4

PROJECT DESCRIPTION

4.1 METHODOLOGY

The project was developed module by module so that each feature could be built, tested on its own and then connected to the rest. The work was done through the following stages.

- **Requirement analysis.** The fragmented patient workflow was studied and divided into appointment, prescription, medicine, reminder, adherence, safety and refill needs.
- **Architecture and data design.** The three-tier structure, the REST interface and the relational entities were decided before coding.
- **Module development.** Appointment scheduling, OCR, medicine management, reminders, adherence analytics, interaction checking and inventory were implemented as separate backend modules with matching frontend screens.
- **Integration.** The modules were linked so that OCR output flows into medicine records, and medicine records feed schedules, interaction checks and inventory.
- **Testing and result capture.** Each workflow was exercised with a real prescription image and screenshots of the results were captured (Figures 4.1 to 4.5).

An incremental approach was suitable because the modules depend on one another in a clear order. Appointments can be built independently, but schedules cannot exist before medicine records, and adherence reports cannot exist before schedules. Building in this order meant that each new module could be tested with real data produced by the previous one [11]. Table 4.1 summarises the phases.

Table 4.1 Development Phases

Phase	Activity	Output
1	Study of the fragmented workflow and review of related work	Problem statement and objectives
2	Design of architecture, data model and interface	Design documents and wireframes
3	Implementation of appointment and medicine modules	Booking dialog, medicine records
4	Implementation of OCR pipeline	Prescription recognition screen
5	Implementation of reminders, adherence, interaction and inventory	Schedule, reports, safety and refill screens
6	Integration and testing with a real prescription	Result screenshots (Figures 4.1 to 4.5)

4.2 SYSTEM MODULES

PillSync consists of seven cooperating modules. Their responsibilities are listed below and described in detail in Section 4.3.

- **Appointment Management** – browsing, slot selection, booking and tracking of appointments.
- **Prescription OCR** – extraction of medicine information from prescription images.
- **Medicine Management** – the central store of the patient's medicines.
- **Scheduling and Reminders** – daily timeline and dose reminders.
- **Medication Adherence** – recording of taken, missed and snoozed doses and analytics.
- **Drug Interaction Detection** – warnings about potentially conflicting medicines.
- **Refill and Inventory Management** – stock monitoring and refill recording.

4.3 DETAILED MODULE DESCRIPTION

4.3.1 Appointment Management

This module handles the interaction between patient and provider. The patient selects a consultation date, sees the time slots that are still available, chooses the appointment type and gives the reason for the consultation. A free-text field allows symptoms or questions to be added, and the booking is finished with a confirmation button. The appointment is then saved in the database with its status, so that scheduled, completed and cancelled appointments can be monitored. Figure 4.1 shows the appointment scheduling screen.

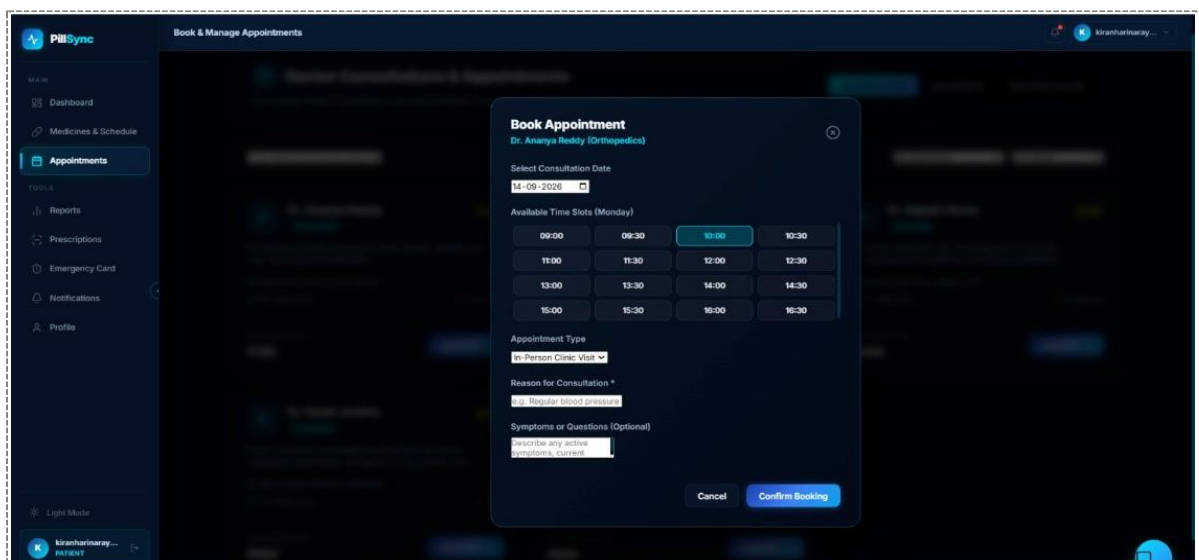


Figure 4.1 Appointment scheduling screen

The module follows the idea of AI-assisted allocation described in Chapter 2: the available slots, together with patient and provider information, are considered when an appointment is allocated. Slots that are already taken are not offered again, which prevents double booking. Table 4.2 lists the status values an appointment can have and what each of them means for the patient.

Table 4.2 Appointment Status Values

Status	Meaning	Patient can
Scheduled	Booked and waiting for the consultation	View details, cancel
Completed	The consultation has taken place	Upload the resulting prescription
Cancelled	The booking was withdrawn	Book a new appointment

The link between this module and the rest of the system is intentional. Once an appointment is marked as completed, the natural next step is to upload the prescription that was issued, and the platform guides the patient in that direction. This is the first point at which the continuity described in Chapter 1 becomes visible.

4.3.2 Prescription OCR

The OCR module converts a prescription image into structured data. The patient uploads or scans the prescription; the image is pre-processed and passed to the OCR engine, and the recognised text is analysed to find the medicine name, generic name, strength, dosage and frequency. The results are displayed in an editable table titled Prescription Recognition, so that the patient can correct any misread value before saving. In the tested example, a prescription from an eye hospital yielded three items, Azopt, Combigan and Xalatan, each with its generic name, strength and frequency. Figure 4.2 shows the OCR result.

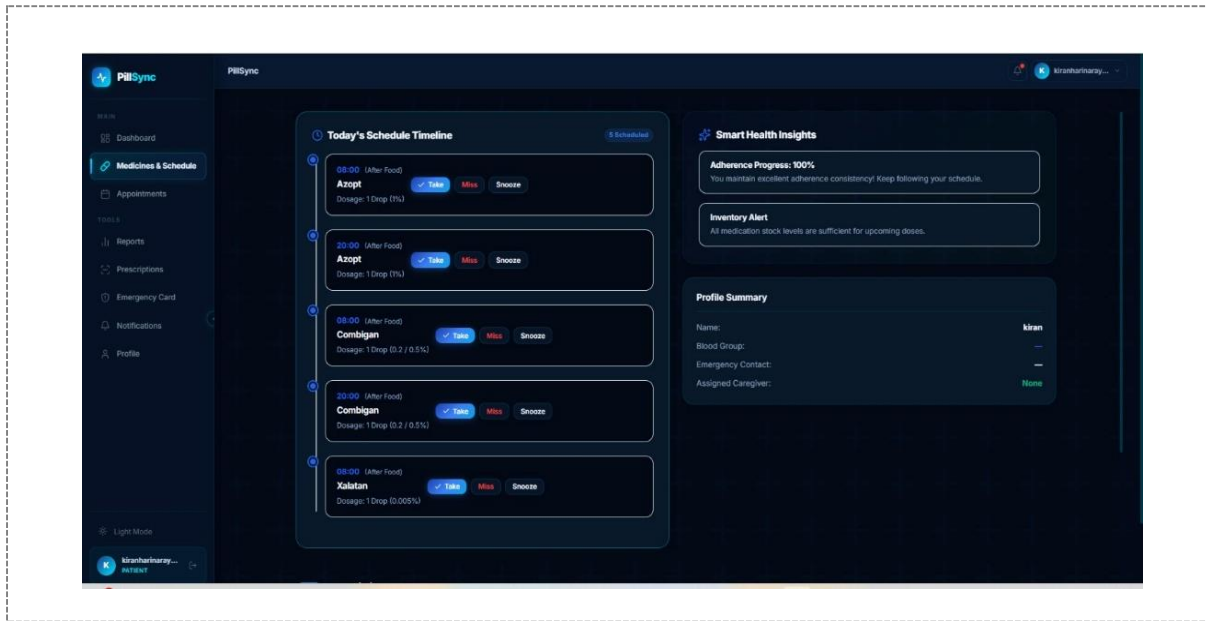


Figure 4.2 OCR result

The pipeline has four automatic stages followed by a manual confirmation. Pre-processing prepares the photograph for recognition, for example by improving contrast and correcting orientation, because handwritten and photographed prescriptions are rarely clean scans [3]. The OCR engine then converts the image into text. A field-identification step separates that text into meaningful pieces, and finally the pieces are laid out as one row per medicine. Table 4.3 lists the fields that are extracted.

Table 4.3 Fields Extracted from a Prescription

Field	Description	Example from the tested prescription
Medicine	Brand name written on the prescription	Azopt, Combigan, Xalatan
Generic name	Active ingredient of the medicine	Shown beside each brand name
Strength	Concentration or amount of the medicine	Shown as a percentage in the table
Dosage	Amount to be taken each time	One drop / one unit as written
Frequency	How often the medicine is taken	Twice daily, once daily
Instructions	Extra directions for use	Entered or corrected by the patient

Keeping the patient in the loop at this point is a safety decision. A wrong dose that quietly turns into a remainder that could cause harm to the patient. By presenting the table and asking the patient to confirm, the system gets the benefit of automation without removing human judgement.

4.3.3 Medicine Management and Reminders

Saved medicines are held as structured records containing name, quantity, dosage, frequency, instructions and scheduled time. Records can be added, viewed, modified and removed. From these records the system builds a Today's Schedule timeline in which every dose is listed with its time and action buttons to mark it as taken, skipped or snoozed. A side panel gives health insights and a short profile summary. Figure 4.3 shows the medicine and reminder dashboard.

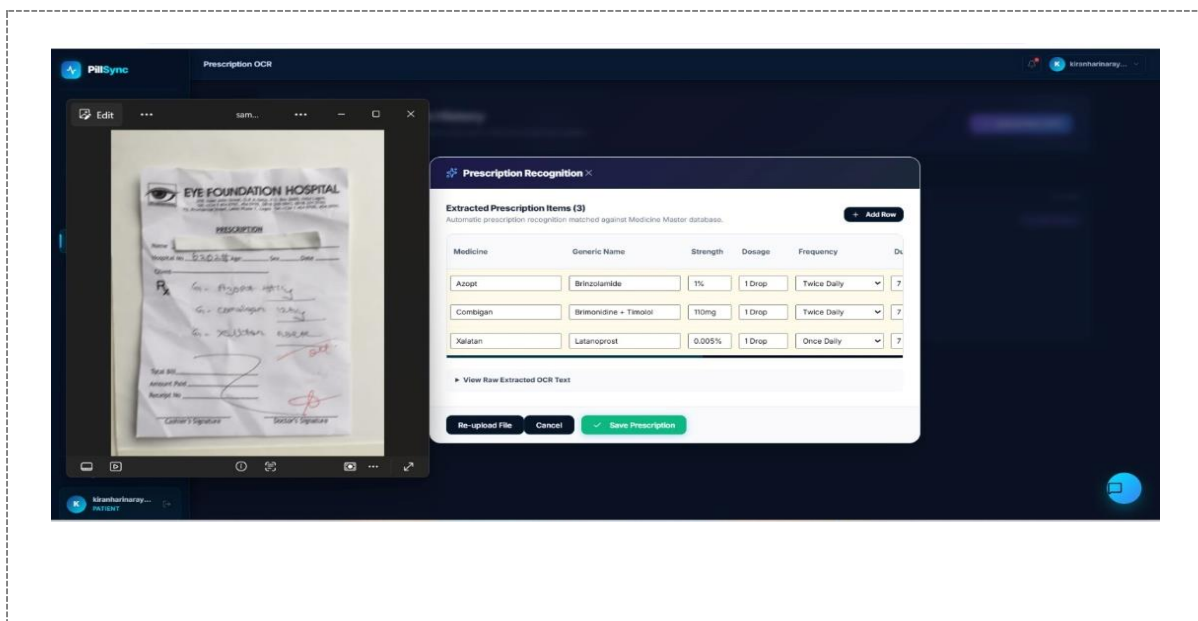


Figure 4.3 Medicine + reminder dashboard

The timeline is the screen the patient is expected to use most often. Doses appear in time order, each with the medicine name and its instruction, and the three action buttons allow the patient to respond with one click. Table 4.4 shows what each action does to the system.

Table 4.4 Dose Actions and Their Effects

Action	Effect on the dose	Effect on analytics
Take	Dose marked as completed	Counted as taken
Skip / miss	Dose marked as not taken	Counted as missed
Snooze	Reminder postponed to a later time	Counted as snoozed until resolved

4.3.4 Medication Adherence Analytics

Every action on a dose is stored as adherence data. The Medication Adherence Report summarises this data through headline figures for overall adherence, a consistency score, the current adherence streak and the total number of doses recorded. A compliance breakdown chart

shows adherence over time, and further panels present medication-safety analysis, an adherence tip and refill predictions. Figure 4.4 shows the adherence analytics screen.

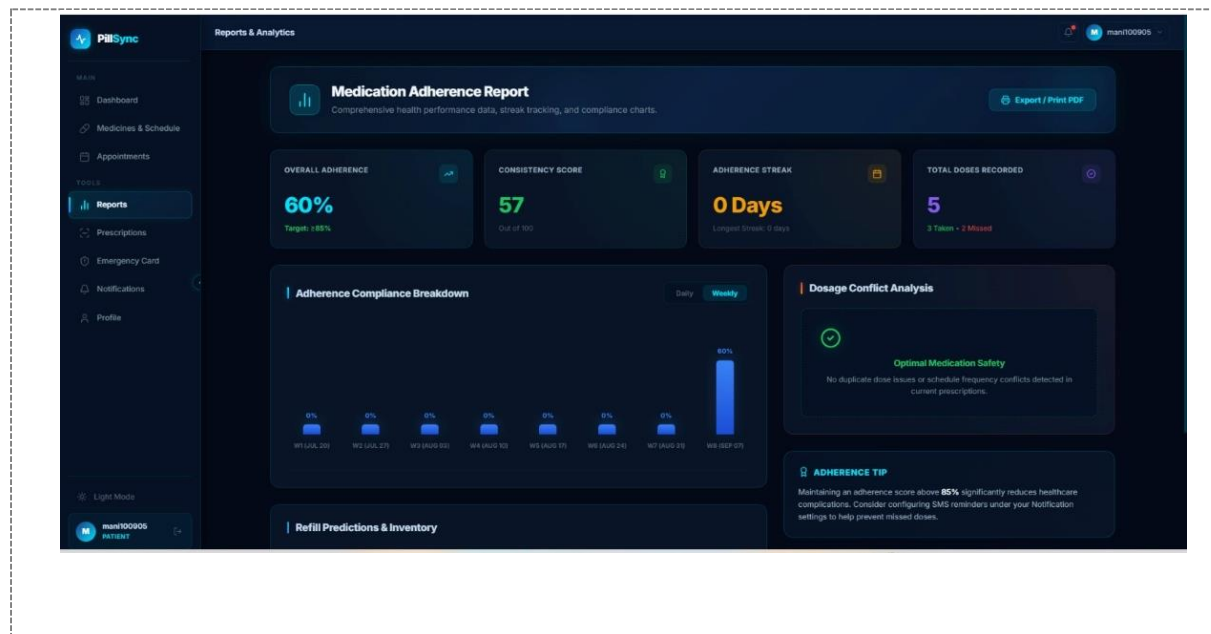


Figure 4.4 Adherence analytics

Adherence is commonly understood as the extent to which a patient's behaviour matches the agreed treatment plan [1], [4]. The report converts the stored dose records into simple figures that a patient can understand without any medical or statistical background. Table 4.5 explains each indicator.

Table 4.5 Adherence Indicators

Indicator	Meaning	How it is read
Total compliance	Percentage of total doses taken Higher value is preferable	meaning no dose was skipped
Stability score	Frequency of doses taken throughout the time	Higher values reflect a more stable dose-taking routine
Adherence streak	Number of consecutive days with all doses taken	Resets when a dose is missed
Total doses recorded	Number of dose records in the period	Shows how much data the figures are based on

As an illustration of the idea, if five doses are recorded and three of them were taken, the overall adherence is three divided by five, or 60 percent. The report shown in Figure 4.4 displays an overall adherence of 60 percent with five doses recorded, which is consistent with this way of reading the figure.

4.3.5 Drug Interaction and Refill Management

The drug-interaction function analyses the combination of medicines stored for a patient. If a potential conflict is found, a warning is displayed; otherwise the patient can continue without alerts. The Inventory and Refills screen lists each medicine with its current stock and a progress bar, summarises how many medicines are low or out of stock, allows filtering by stock state and provides a Record Refill button for each medicine. Figure 4.5 shows the interaction and refill result.

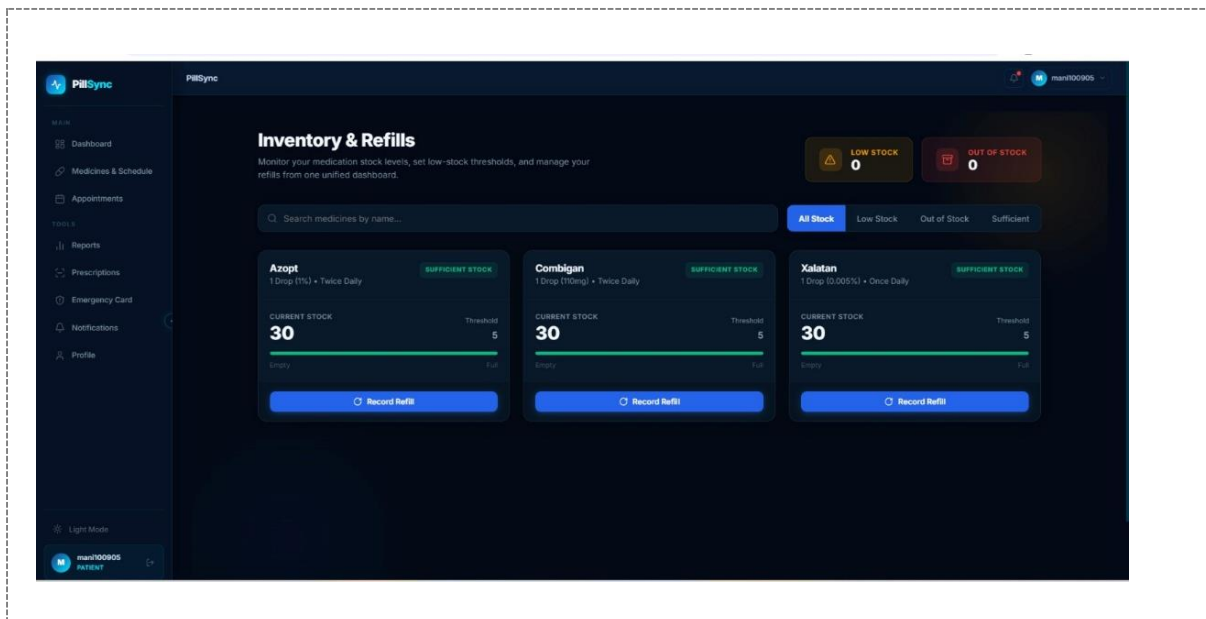


Figure 4.5 Drug interaction + refill result

Two small tables describe how this module treats its results. Table 4.6 shows the way interaction analysis is presented, and Table 4.7 lists the stock states used on the Inventory and Refills screen.

Table 4.6 Handling of Interaction Results

Result	What the patient sees	What the system does
No conflict found	Medication-safety panel shows an optimal status	Continues normally
Potential conflict found	A warning that names the combination	Displays the warning; leaves the decision to the patient and clinician

Table 4.7 Stock States in Inventory Management

State	Meaning	Action offered
Sufficient stock	Quantity comfortably above the need	None required
Low stock	Quantity is close to running out	Record refill
Out of stock	No quantity left	Record refill urgently

4.3.6 Integration Between Modules

The modules are connected rather than independent. The appointment leads to a consultation and a prescription; the prescription becomes medicine records through OCR; the medicine records drive scheduling, interaction checking and inventory; and dose responses flow into adherence analytics. This chain is the defining feature of the system and the reason the patient can manage the whole journey from one interface.

4.4 LOGIC USED IN THE MODULES

This section describes the main rules used inside the modules in plain language. They are presented as design logic and are not a copy of the source code.

4.4.1 Slot Availability Logic

When the patient chooses a date, the system lists the slots of that day and removes those that have already been booked. Only the remaining slots are shown in the grid of the booking dialog. When the booking is confirmed, the chosen slot is stored with the appointment so that it cannot be offered to anyone else.

4.4.2 Field Identification Logic

After text extraction, each line of the prescription is examined for patterns that indicate a medicine entry: a medicine name, a number followed by a unit or percentage sign, and words or abbreviations that describe frequency. Lines that contain such patterns are grouped into a row, and the individual fields are assigned to the columns of the recognition table. Anything the system cannot assign with confidence is left for the patient to complete.

4.4.3 Adherence Calculation

The complete adherence is figured out through division of the number of doses taken by the number of doses noted and applied in the percentage form. The string represents the number of uninterrupted days, including the current date, with no dose missing. The compliance diagram gathers the same information, analyzing it daily or weekly, so that one can see how the treatment process has progressed over time.

4.4.4 Refill Prediction Logic

For each medicine, the number of days of supply left is estimated by dividing the current quantity by the quantity consumed per day, which follows from dosage and frequency. For example, 30 units used at two units per day last about 15 days. When this estimate falls below a threshold, the medicine is flagged as low stock and the refill button is emphasised. The refill prediction panel in the reports screen presents this information together with the inventory.

4.4.5 Input Validation

Data entered by the patient or recovered by OCR is checked before it is saved. Table 4.8 lists the main validation rules.

Table 4.8 Input Validation Rules

Input	Rule
Appointment date	Must be a valid date and offer at least one free slot
Appointment reason	Required field, cannot be empty
Prescription image	Must be a supported image file
Medicine name	Required, cannot be blank
Quantity	Must be a non-negative number
Frequency and time	Must be in a form the scheduler can interpret

CHAPTER 5

IMPLEMENTATION AND RESULT DISCUSSION

This chapter explains how the design was implemented, describes the environment in which the system was tested and presents the results with the help of the screenshots captured during testing. It also discusses what the results show and where the work is limited.

5.1 IMPLEMENTATION

5.1.1 Frontend Implementation

The **frontend** is a React application with a dark, dashboard-style interface. A side navigation bar gives access to the dashboard, medicines and schedule, appointments, reports, prescriptions, emergency card, notifications and profile sections. Forms, dialogs and charts are rendered on the client and exchange data with the backend through REST calls [7]. React's component model suits this interface well, because cards such as a medicine card or a dose card are repeated many times with different data.

5.1.2 Backend Implementation

The **backend** is written with FastAPI. It receives the frontend requests, validates them, runs the application logic of each module and communicates with PostgreSQL [6]. Separating the modules inside the backend keeps the appointment, OCR, medicine, reminder and adherence logic independent while still allowing them to share data. FastAPI also generates interactive documentation of the operations it exposes, which is convenient while testing the endpoints.

5.1.3 Database Implementation

PostgreSQL stores users, appointments, prescriptions, medicines and adherence data in related tables [8]. Foreign keys link each child record to its parent, so a medicine always belongs to a prescription and a dose record always belongs to a medicine. Queries for the adherence report group dose records by period and status, and queries for the inventory screen read the quantity stored in each medicine record.

5.1.4 OCR Implementation

The OCR pipeline takes in a prescription image, does pre-processing, extracts the text, and identifies the name, strength, dosage and frequency of medicine. The results are shown in an editable table and only upon approval does it become a set of records, putting the patient in charge of the completed data.

5.1.5 Reminders, Adherence, Interaction and Inventory

Reminders and adherence are derived from the stored schedule. A dose is displayed at its scheduled time; the patient's response is stored as taken, missed or snoozed, and aggregate statistics are computed from these records. **Interaction checking** inspects the medicine list and raises a warning when a potential conflict is found, while the **inventory** module keeps the current quantity of every medicine and highlights low or empty stock.

5.2 EXPERIMENTAL SETUP

The system was tested on a laptop using a modern web browser. A demonstration patient account was created and a real, photographed prescription from an eye hospital was used as the test input. The prescription contained three medicines, which were carried through every stage of the workflow. Table 5.1 summarises the configuration.

Table 5.1 Test Environment Configuration

Item	Configuration
Application	PillSync web application
Frontend / Backend / Database	React / FastAPI / PostgreSQL
Client	Modern web browser on a laptop
Test account	A demonstration patient profile
Test input	A photographed prescription from an eye hospital containing three medicines
Output captured	Screenshots of each module (Figures 4.1 to 4.5)

5.3 TEST SCENARIOS

Seven functional scenarios were prepared, one for each major capability. They follow the order of the patient workflow so that the output of one scenario becomes the input of the next. The scenarios and the behaviour expected from the system are shown in Table 5.2.

Table 5.2 Functional Test Scenarios

No.	Scenario	Expected Behaviour
1	Open the booking dialog, choose a date and slot, select type and reason, confirm	Appointment is accepted and stored
2	Upload a prescription image	Medicine name, strength, dosage and frequency appear in an editable table
3	Save the recognised table	Medicine records are created and appear in the schedule
4	Mark doses as taken, skipped or snoozed	Status is recorded against the dose
5	Open the adherence report	Overall adherence, consistency, streak and dose count are shown

No.	Scenario	Expected Behaviour
6	Open inventory and refills	Current stock per medicine and low / out-of-stock counts are shown
7	Review the medication-safety panel	Potential conflicts are flagged, or no-conflict status is shown

5.3.1 Detailed Test Cases

For each scenario the main inputs, steps and pass criteria are listed in Table 5.3. A scenario is considered passed when the observed behaviour matches the pass criterion and the corresponding screen was captured as evidence.

Table 5.3 Detailed Test Cases

Test case	Input / Steps	Pass criterion	Evidence
TC1 Booking	Select date, slot, type, reason; press Confirm Booking	Booking accepted without error	Figure 4.1
TC2 OCR	Upload prescription photograph	Three medicines listed in the recognition table	Figure 4.2
TC3 Save records	Press Save Prescription after review	Records visible in medicines and schedule	Figure 4.3
TC4 Dose action	Press Take, Skip or Snooze on a dose	Status stored and reflected in analytics	Figures 4.3, 4.4
TC5 Adherence	Open Reports and Analytics	Figures and chart displayed	Figure 4.4
TC6 Inventory	Open Inventory and Refills	Stock per medicine and counts displayed	Figure 4.5
TC7 Safety	Open medication-safety panel	Status or warning displayed	Figures 4.4, 4.5

5.4 RESULTS

The screenshots captured during testing confirm that the main workflow works from end to end.

- **Appointment schedule (Figure 4.1).** The booking dialog offered a date selector, a grid of time slots, an appointment-type field, a reason field and an optional symptoms box, with Cancel and Confirm Booking controls.
- **OCR (Figure 4.2).** The prescription image was read and three medicines were extracted with generic name, strength, dosage and frequency, ready for review and saving.
- **Schedule and reminders (Figure 4.3).** The Today's Schedule timeline listed the doses with Take, Skip and Snooze actions next to each one.

- **Adherence (Figure 4.4).** The report displayed an overall adherence of 60%, a consistency score of 57, an adherence streak of 0 days and 5 doses recorded, together with a compliance breakdown chart.
- **Inventory (Figure 4.5).** All three medicines showed a current stock of 30 units, with zero low-stock and zero out-of-stock items and a Record Refill button for each.

Table 5.4 Feature Validation Summary

Feature	Evidence	Observation
Appointment booking	Figure 4.1	Booking form with slot selection displayed
Prescription OCR	Figure 4.2	Three medicines extracted into structured form
Medicine schedule and reminders	Figure 4.3	Daily timeline with dose actions
Adherence analytics	Figure 4.4	Adherence statistics and chart generated
Inventory and refill	Figure 4.5	Stock levels and refill controls shown
Drug-interaction warning	Figures 4.4 and 4.5	Medication-safety status displayed

5.5 RESULT ANALYSIS

5.5.1 Appointment Schedule

The booking dialog gathers all the details you need for an appointment all on one screen. The patient does not have to move between pages, and the grid of time slots makes the available options visible immediately. The result confirms that the first stage of the workflow works and that appointment data can be stored for later monitoring.

5.5.2 Prescription Recognition

The main finding is that a handwritten prescription originally in the form of a photo has transformed into a table containing structured pharmaceutical entries. This enables the elimination of one of the most stubborn and troublesome manual steps in the traditional procedure. The demonstrated table indicates that the system considers the outcome of OCR as a suggestion instead of the definite answer, which is suitable for medical information.

5.5.3 Schedule and Reminders

Once the medicines had been saved, the schedule appeared without any additional data entry. The timeline presents each dose with its time, and the action buttons allow the patient to respond immediately. This demonstrates the benefit of linking the OCR module to the medicine module: the patient typed no medicine details at all.

5.5.4 Adherence Analytics

The adherence report converts raw dose responses into figures that are easy to interpret. The overall adherence of 60 percent, the consistency score and the streak give the patient a quick sense of how well the routine is being followed. The values in the screenshot reflect the small amount of test data recorded during the demonstration and should not be read as a clinical result.

5.5.5 Inventory and Refills

The inventory screen shows the quantity of each medicine as a number and a progress bar and counts how many medicines are low or out of stock. In the test, all three medicines had adequate stock, so no low-stock alerts appeared. The Record Refill button is available for every medicine so that replenishment can be logged as soon as it happens.

5.5.6 Medication Safety

The medication-safety information is presented alongside the adherence and inventory results. For the tested medicines, the interface displayed an optimal status, meaning that no conflict was reported. The function is designed to show a warning whenever a potential interaction is found in the stored medicine list.

5.6 COMPARISON WITH THE CONVENTIONAL APPROACH

The value of the integrated platform becomes clearer when it is compared with the conventional approach in which separate tools are used. Table 5.5 compares the effort needed for the main tasks of the patient journey.

Table 5.5 Conventional Approach Compared with PillSync

Task	Conventional approach	PillSync
Booking a consultation	Separate portal or telephone call	Booking dialog inside the platform
Recording prescription details	Manual typing into another application	OCR extraction followed by review
Setting reminders	Manual entry of each medicine and time	Generated from saved medicine records
Tracking adherence	Memory or handwritten notes	Automatic from dose actions
Checking interactions	Separate lookup tool	Automatic analysis of the stored medicine list
Monitoring stock	Manual counting	Inventory screen with low-stock flags

5.7 DISCUSSION

The results show that the main benefit of the system comes from reuse of data. The patient typed nothing about the medicines: the OCR result produced the records, the records

produced the schedule, and the schedule produced the adherence report and the inventory view. In the conventional approach each of these steps would need a separate manual entry.

The design also keeps the patient in charge. OCR output is presented for verification instead of being saved silently, reminders require a response, and interaction results are presented as warnings rather than automatic changes. This is important in a healthcare setting where a wrong automatic decision can be harmful.

A further observation is that the quality of every later stage depends on the first digitization step. If a medicine name or frequency is misread and not corrected, the schedule, the stock estimate and the interaction check will all inherit the mistake. This is why the review table is a necessary part of the workflow and not an optional extra. Improving the recognition accuracy is therefore the most valuable direction for further work.

5.8 LIMITATIONS

- OCR accuracy depends on the quality and legibility of the prescription image; badly written or blurred prescriptions may need manual correction.
- The project was demonstrated with a small sample of test data, and no large-scale quantitative accuracy evaluation has been carried out.
- Interaction warnings are only as complete as the interaction knowledge they rely on and should never replace the advice of a doctor or pharmacist.
- The system does not yet integrate with hospital records or pharmacy ordering services.
- Reminders are shown inside the application; delivery through external channels such as SMS has not been implemented.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

This project presented PillSync, an intelligent healthcare management system that connects appointment scheduling with prescription digitization and continuous medicine management. Through a React frontend, a FastAPI backend and a PostgreSQL database, the platform lets a patient book an appointment, convert a prescription image into structured medicine records with OCR, receive dose reminders, record taken, missed and snoozed doses, view adherence analytics, obtain interaction warnings and keep track of inventory and refills.

The implemented screens demonstrate that the stages of the patient journey can be linked so that data created once is reused everywhere. The work therefore reduces manual effort and offers a more organised and continuous healthcare experience than the use of separate tools.

The objectives set in Chapter 1 have been met at the level of a working prototype. Appointments can be booked and stored, prescriptions can be digitized with OCR, medicines can be managed and scheduled, doses can be tracked, adherence can be analysed, interaction warnings are available and stock can be monitored. What remains is the step from a demonstrated prototype to a deployed service, which requires larger-scale testing, stronger security and integration with the wider healthcare ecosystem.

6.2 FUTURE WORK

- Improve OCR accuracy for handwritten prescriptions through better image pre-processing and trained recognition models.
- Add multilingual prescription support.
- Join up with pharmacies with regard to placing orders for refills and cooperate with hospitals for prescription transfer.
- Provide reminders through mobile push notifications, SMS or e-mail.
- Widen the drug interactions module with an even bigger reference database that is updated regularly . Add caregiver and doctor views so that family members and clinicians can follow adherence with the patient's consent.
- Conduct a structured evaluation with real users and measure OCR accuracy and adherence improvement.

Table 6.1 groups these enhancements by expected effort and benefit, which may help in deciding the order in which they should be taken up.

Table 6.1 Proposed Enhancements

Enhancement	Main benefit	Relative effort
Better handwriting recognition	Fewer manual corrections after OCR	High
Multilingual support	Wider accessibility for patients	Medium
External reminder channels	Reminders reach the patient outside the application	Medium
Larger interaction knowledge base	More complete safety warnings	Medium
Pharmacy and hospital integration	Less manual transfer of data, direct refills	High
Caregiver and doctor views	Shared monitoring of adherence	Medium
User study and accuracy evaluation	Evidence of benefit and reliability	Medium

6.3 PRACTICAL IMPACT

A platform of this kind is most useful for people who must manage several medicines over a long time, such as elderly patients and those with chronic conditions, and for family members who help them. By reducing the number of separate tools and the amount of typing, it lowers the effort of staying on treatment. Better visibility of adherence and stock may also help patients and clinicians discuss problems earlier. These benefits are expected consequences of the design and would need to be confirmed through a formal study with real users [2], [12].

APPENDIX 1 - DATA MODEL AND STATUS RULES

Table A.1 lists the status values used by the main modules.

Table A.1 Status Values Used in the System

Module	Status	Meaning
Appointment	Scheduled	Booked and awaiting the consultation
Appointment	Completed	Consultation has taken place
Appointment	Cancelled	Booking withdrawn
Dose	Taken	Patient confirmed the dose; counts towards adherence
Dose	Missed / Skipped	Dose not taken; reduces adherence
Dose	Snoozed	Dose postponed; reminder raised again later
Inventory	Sufficient	Stock adequate for the planned doses
Inventory	Low stock	Quantity close to running out; refill advised
Inventory	Out of stock	No quantity remaining; refill required

Table A.2 shows the structure of a medicine record as it is produced from the OCR result. The values are taken from the tested prescription as read from the recognition screen and should be verified against the original image before the report is submitted.

Table A.2 Sample Structured Medicine Records

Medicine	Generic name	Strength	Frequency
Azopt	Brinzolamide	1%	Twice daily
Combigan	Brimonidine + Timolol	As on prescription	Twice daily
Xalatan	Latanoprost	0.005%	Once daily

APPENDIX 2 - DETAILED TEST CHECKLIST

Table B.1 End-to-End Project Verification Checklist

Item	Check	Status
Appointment booking	Date, slot, type and reason accepted	Demonstrated (Fig. 4.1)
Prescription upload	Image accepted by OCR module	Demonstrated (Fig. 4.2)
OCR extraction	Name, strength, dosage and frequency returned	Demonstrated (Fig. 4.2)
Medicine records	Records created from OCR output	Demonstrated (Fig. 4.3)
Schedule and reminders	Daily timeline with dose actions	Demonstrated (Fig. 4.3)
Adherence analytics	Statistics and chart generated	Demonstrated (Fig. 4.4)
Inventory and refill	Stock and refill buttons displayed	Demonstrated (Fig. 4.5)
Interaction warning	Warning shown when a conflict exists	Status panel shown (Fig. 4.4)

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